

Thematic Digital Competence Centers

Application form

1. General information

1a. Project title

SYNOPSIS: Synergy Network and Platform for Integrating Audiovisual Data Archiving and Stewardship in Social Sciences and Humanities.

1b. Relevant Thematic DCC Domain

Please select the relevant domain.

- ☐ Life Sciences and Health (LSH)
☐ Natural and Engineering Sciences (NES)
☒ Social Sciences and Humanities (SSH)

1c. Project duration

24 months.

1d. Responsible institution(s) and department

Organisatie van aanvraag:

- *Radboud University, Faculty of Arts and Centre for Language Studies*

Organisaties van uitvoering:

- *Radboud University, ILS, Digital Competence Centre*
- *Radboud University, Donders Institute for Brain and Cognition*
- *University of Amsterdam (UvA), Institute for Logic, Language & Computation (ILLC)*
- *Vrije Universiteit Amsterdam, Faculty of Humanities, Language and Communication, and Network Institute*
- *Erasmus University Rotterdam, Erasmus School of Social and Behavioural Sciences, DRIFT.*
- *Hasso Plattner Institute, Potsdam, Germany*
- *DANS*
- *SURF (Collaborative Organization for ICT in Dutch Education and Research)***
- *E-Science Center***
- *ODISSEI (Open Data Infrastructure for Social Science and Economic Innovations)***

***Consultations added in this revision/ future collaboration to be explored*

1e. Lead applicant and co-applicants

Note. While one of us submits (much like a corresponding author in academic publishing), this is a team science project and the primary role lead applicant and all co-applicants share in the common language we've adopted for this project is of co-pilot.

Title, first name, initials, surname	Affiliation	Department	Role(s)
<i>Mark Dingemanse</i>	<i>RU</i>	<i>Centre for Language Studies</i>	<i>Lead applicant, co-pilot</i>
<i>Babajide Owoyele</i>	<i>EUR & HPI</i>	<i>DRIFT</i>	<i>Co-lead applicant, Co-pilot</i>
<i>Wim Pouw</i>	<i>RU</i>	<i>FSW/Donders</i>	<i>Co-pilot</i>
<i>Bogdana Huma</i>	<i>VU</i>	<i>Language, Literature, and Communication</i>	<i>Co-pilot</i>
<i>Henk van de Heuvel</i>	<i>RU</i>	<i>Humanities Lab</i>	<i>Co-pilot</i>
<i>Inge Slouwerhof</i>	<i>RU</i>	<i>Information and</i>	<i>Co-pilot</i>

		Library Services - DCC	
James Trujillo	UvA	ILLC	Co-pilot

1f. Keywords

Audiovisual Data, Multimodal Analysis, Digital Competence, Privacy-aware Archiving, Social Sciences

1g. Disciplines

	Discipline:
Main discipline:	37.10.00 Software for humanities
Other disciplines (if applicable)	37.90.00 Computers and the humanities, other
	16.60.00 Artificial intelligence, expert systems
	Select an item.

1h. Abstract (max. 500 words)

SSH researchers increasingly rely on audiovisual data to study complex human behaviours and social interactions, but current practices and systems for handling such data lag behind. Privacy is the main challenge to making research reproducible and data reusable. We aim to increase audiovisual data reusability while protecting privacy by integrating masking technology into research workflows. Masking tools allow researchers to hide faces, voices, and other identifying traits while enriching the data with de-identified body movement data and other social signals. The goal is to strike a balance between removing identifiable information and increasing the reusability and interpretability of the data. We do this by co-designing new standards, practices, and workflows with stakeholders, ensuring that audiovisual data in SSH fields can be “as open as possible and as closed as necessary.”

Researchers and support staff tell us they are hesitant to adopt technological advances in audiovisual masking and perceive them to be too programming-heavy or hard to fit into current workflows. Directly addressing this, SYNOPSIS aims to provide an easy-to-use, open-source platform with tools for protecting privacy and clear guidelines and training. By helping researchers better manage sensitive data, the project will encourage more responsible and widespread use of audiovisual data in research. SYNOPSIS integrates and extends an existing user-friendly tool called MaskAnyone, which allows researchers to hide sensitive information using a point-and-click user interface. MaskAnyone’s modular design will enable researchers to choose and set up additional data-enriching tools for handling video, audio, or movement data.

The SYNOPSIS project has two main parts: building a platform and training researchers to use it. Platform development focuses on creating a reliable and easy-to-navigate cloud-based system for storing and sharing audiovisual data in cooperation with partners like SURF and DANS. Tools are supported by video tutorials, FAQs, and workflows (e.g., informed consent templates). On the training side, we work with four universities in the Netherlands to upskill researchers and technical staff. Our Masking School will train researchers and data stewards in masking, archiving, and sharing masked video and audio data, ensuring strong digital competence and institutional support for masking practices in SSH. Currently, staff in the Netherlands are not trained in these areas.

In collaboration with partners like SURF and DANS and RIs like ODISSEI and CLARIAH, our Masking Lab will address more advanced challenges in platform building and training. Here, we identify lacunae in current tools and chart the risks associated with particular forms of masking. A Hackathon-style benchmark event and study and a pilot certification program will help lay the foundations for building a community to navigate pressing risks.

With the SYNOPSIS platform and training track, we want to build long-term solutions and support infrastructure for masking audiovisual data that meet FAIR guidelines and prepare for the future of maximally transparent, reproducible, and privacy-aware research workflows in SSH. Our training and dissemination methods, coupled with our strong network of partner organizations, will ensure community-wide awareness and a lasting impact on the SSH research landscape.

1i. Public project title and public summary

NL

SYNOPSIS : Synergie Netwerk en Platform voor Integratie van Audiovisuele Data-analyse en Archivering in

de Sociale Wetenschappen

SYNAPSIS is een samenwerkingsproject gericht op het verbeteren van het beheer, delen en analyseren van audiovisuele data in de Sociale Wetenschappen en Geesteswetenschappen. Het project biedt een veilig, privacy-beschermend platform dat geavanceerde masking-tools integreert voor audiovisuele data, waardoor onderzoekers gevoelige informatie kunnen beschermen en tegelijkertijd data toegankelijker maken voor analyse. Naast het platform worden uitgebreide trainingsprogramma's aangeboden om onderzoekers en data-stewards de digitale vaardigheden te bieden die nodig zijn om masking en multimodale analyses te integreren in hun workflows, wat open wetenschap bevordert en de transparantie van onderzoek verbetert.

ENG
SYNAPSIS: Synergy Network and Platform for Integrating Audiovisual Data Archiving and Stewardship in Social Sciences and Humanities

SYNAPSIS is a collaborative project advancing audiovisual data management, sharing, and analysis in the Social Sciences and Humanities (SSH). The project provides a secure, privacy-aware platform that integrates cutting-edge masking tools for audiovisual data, allowing researchers to protect sensitive information while making data more accessible for analysis. Alongside the platform, comprehensive training programs will equip researchers and data stewards with the skills to implement masking and multimodal analytics in their workflows, promoting open science and improving research transparency.

2. Project information

2.1 Involvement of TDCC domain in drafting application (max. 400 words)

SYNOPSIS is shaped by an inclusive, iterative process involving the full spectrum of potential users, support staff, support infrastructure, and future developers and local policymakers. The applicants directly involved SSH research institutions and digital infrastructure providers, including DANS, SURF, Radboud Humanities Lab, and CLARIN/CLARIAH. Researchers and support staff at several Donders Network Institutes, and Summer Schools held at Radboud, Potsdam, Göttingen, and Poland, helped us refine the SYNOPSIS project goals, which includes proper integration of archiving workflows and maximal reuse of digital infrastructure (SURF SANE). Radboud University's Humanities Lab will develop the platform's front-end and also provides important links to DANS and SURF infrastructure. The involvement of the DCC coordinator as a co-pilot ensures local synergies within and outside RU.

During the iterative review process the proposal was further shaped by participants from the NWO OSF project "[Building a community of practice for open naturally occurring data](#)" as well as by interactions with RDM experts and SSH stakeholders, including Esther Plomp (UU), Jetze Toubert (Ghent), Ricarda Braukmann (DANS) and Eduard Klapwijk (Leiden). A first set of pilots at Radboud University helped to scope synopsis as a pilot for GDPR-compliant integration of data anonymization at scale and increased the likelihood of successful integration in centralized digital infrastructures. Further, feedback from Data Stewards and experts from SURF, RU, VU, and the NL eScience Center nudged us to open up our masking practices catalog to incorporate additional tools, such as penciling and DeepPrivacy2. Reviews from [Andrea Scharnhorst](#), [Rene Bekkers](#), [Roeland Ordeman](#) and [Ahmad Hesam](#) led to a crucial rethinking and redesign of the SURF pilot implementation. Feedback we elicited from the RU Social Science Lab helped us broaden to focus of the Masking Lab to include ethics, FAIR principles, and metadata. The [Data Stewards Interest Group](#) community will help disseminate our workshops and surveys and invite participants to our workshops. We concluded the grant writing phase with in-depth discussions with the NL eScience Center on the project's sustainability. Additionally, RU Data Ethics Coordinator [Miriam Kos](#) connected us to Radboud Teacher Academy pilot/use case with [Robin Satter](#) now on board as advisory.

We are incredibly proud of the resulting proposal. SYNOPSIS addresses the real needs of the SSH community while advancing critical areas in augmenting data stewardship, privacy infrastructure, network building and FAIRification. If funded, it will mark a substantial advance for sustainable digital competence in the SSH domain.

2.2 Roadmap ambition/activity cluster/challenge

Which roadmap ambition/cluster/challenge does this project aim at? Make a selection within the domain marked under 1b.

NES	SSH	LSH
☐ FAIR data	☒ FAIRification of data and software	☐ Strengthen good data stewardship and harmonise data access practises
☐ Sustainable software and e-science	☒ Raising awareness	☐ Enhance the interoperability of digital solutions and resources
☐ Connections to the international activities	☐ Knowledge enhancement	☐ Strengthen the capacity and expertise base in digital research
☐ Long-term data archiving	☐ Addressing pressing issues	
☐ Locate computing capacity close to the storage	☒ Building a network	
☐ Human capital		
☐ Cross-field collaborations		

2.3 Relevance (max. 400 words)

As the use of rich audiovisual data in SSH research continues to grow, so does the need for effective data management strategies. A recent policy paper argues that audio and video recordings should not be destroyed, overturning long-held beliefs in SSH fields ([Resnik et al. 2024](#)). SYNAPSIS 1.0 aims to answer to louder calls for augmenting audiovisual data management by advancing the digital competence data stewardship and applies FAIR (Findable, Accessible, Interoperable, Reusable) principles, which are central goals of the TDCC Social Sciences and Humanities (SSH) roadmaps.

SYNAPSIS innovates on existing digital infrastructures (e.g., MaskAnyone, SURF), ensuring relevant expertise (e.g., SURF has been working on secure video conferencing software, a highly pertinent project for SYNAPSIS). SYNAPSIS is also a practical solution for researchers as it offers a user-friendly platform for securely managing and analyzing audiovisual data. SYNAPSIS will provide a secure data portal with transparent processing of audiovisual data by open-source software tools, combined with appropriate data governance measures conveyed to data stewards. These are essential prerequisites to ensure that masking becomes a viable alternative to audiovisual data removal by default.

Pilot Use Cases and Sustainability Plan via Grants

We're collaborating with Radboud Teachers Academy (RTA) to implement the MaskAnyone Toolkit, enabling GDPR-compliant de-identification in teacher training assessments while preserving trainee visibility. RTA's trainees record videos in real-world classrooms, which serve as essential learning tools but require privacy protections when repurposed for broader educational and research use. By anonymizing pupils through face-swapping and voice alteration while keeping teachers identifiable with consent, MaskAnyone supports privacy-compliant video sharing. This pilot will also involve RTA's privacy officer and ethical committee, facilitating compliance assessments to lay the groundwork for potential adoption across Radboud University. Additionally, in partnership with Joaquin Santuber at Johannes Kepler Universität Linz, we're exploring methods to de-identify Chilean court videos for research. The sustainability of SYNAPSIS is supported by partnerships with SURF and NLeSC for secure, scalable cloud operations, alongside "train-the-trainer" workshops to build local expertise.

We are exploring synergies with DARIAH Campus to ensure SYNAPSIS's long-term relevance. Through DARIAH's "Hosted Resource" option, SYNAPSIS training materials could be preserved sustainably and accessible to users beyond the project's lifecycle. This partnership would offer SYNAPSIS ongoing visibility within the Social Sciences and Humanities research community, with cataloguing opportunities in the SSH Open Marketplace further extending its reach and usability.

2.4 Project outline including work packages (max. 1600 words)

Our project aims to innovate masking tools and infrastructure to provide workflows and training materials for users and support staff while publishing all training, code, and adapted workflows on a centralized platform open for reuse by other institutions (both within and outside NL).

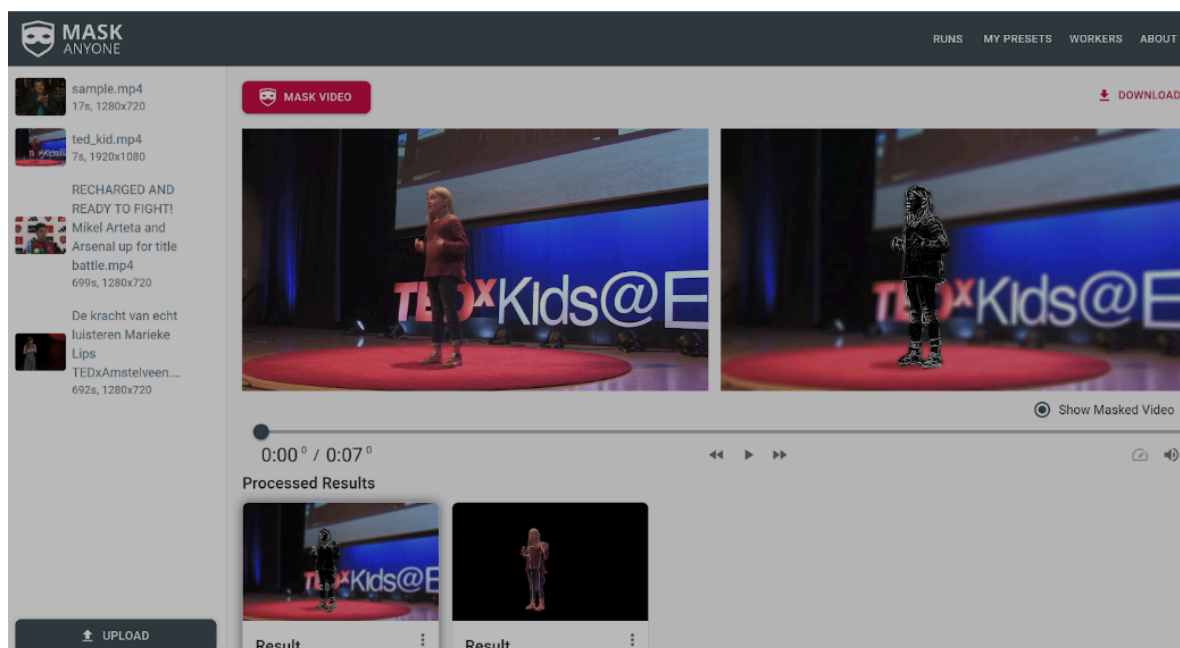


Figure 1a The MaskAnyone Toolkit User Interface see (<https://github.com/MaskAnyone/MaskAnyone>)

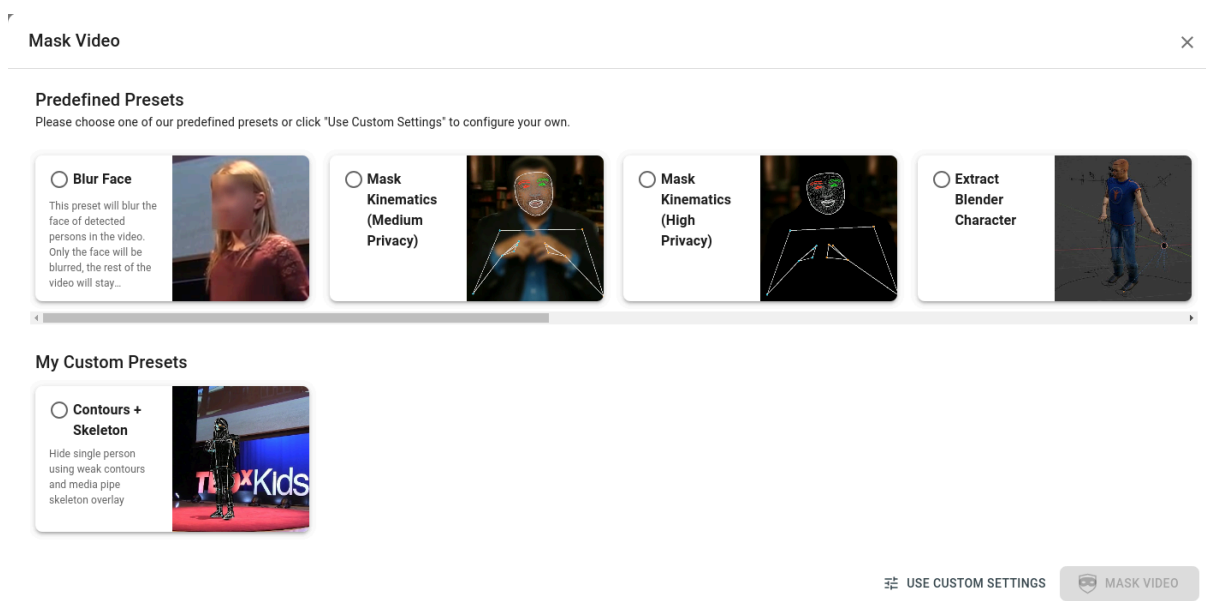


Figure 1b The MaskAnyone Toolkit User Interface see (<https://github.com/MaskAnyone/MaskAnyone>)

Work Package 1: Platform Foundation and Community Engagement (Months 1-12)

WP 1 Overview:

Track	Component	Target Audience	Who	What	Deliverables
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Training Track	<i>Masking School</i>	<i>PIs, support staff (~N=20), ECRs, data stewards (~N=30)</i>	<i>Trujillo, Owoyele, Pouw</i>	<i>Two live/hybrid 1.5-hour sessions at VU, two live/hybrid 4-hour sessions at UvA & Rotterdam, 3-day summer/winter schools at Radboud</i>	<i>Training materials, video tutorials, participant certification</i>
Platform Track	<i>SYNOPSIS Pilot Testing</i>	<i>Radboud University, SURF</i>	<i>Dingemanse, van de Heuvel</i>	<i>Pilot deployment at Radboud, focusing on performance and GDPR compliance</i>	<i>Completed pilot, technical report</i>

Objective: Establish the foundation for the SYNOPSIS platform and facilitate the incorporation of masking into the workflows of the SSH research community.

Platform Track:

The primary focus will be transforming the MaskAnyone toolkit (see Figure 1a and b above) into a user-friendly platform supporting secure file exchange and adaptable workflows. Collaboration with **Mark Dingemanse** and **Henk van de Heuvel** at **Radboud University's Digital Humanities Lab** will ensure the necessary technical infrastructure for institutional and national use, including integration with **SURF**. The Masking toolkit pilot will be implemented for the Radboud Teachers Academy with support from one research coordinator and a privacy officer ([Robin Satter](#)). We will integrate with 1) their existing video-based assessments of teacher trainees and 2) open science research practices.

Concrete actions include:

- Developing secure infrastructure for GDPR-compliant file storage, seamless file exchange, and processing.
- Integrating access to the masking tools for de-identification of sensitive audiovisual data.
- Piloting the platform in real-world research environments at **Radboud** to ensure platform performance and adaptability.

Overview of Constructive Alignment for Masking School

Component	Details
Teaching and Learning Activities (TLA)	- Introduction and theoretical background on masking tools. - Demonstrations of MaskAnyone and related tools. - Hands-on exercises to practice audiovisual data masking. - Discussions and breakout sessions for sharing challenges and best practices.
Assessment	- Completion of exercises: Participants will demonstrate data masking skills. - Case studies: Discussion of ethical and practical scenarios. - Feedback from instructors: Formative assessments during sessions. - Certificate of completion for successful participation.

Course Objectives	- Equip participants with practical skills for audiovisual data masking. - Ensure understanding of ethical frameworks and GDPR compliance. - Promote FAIR principles for data management. - Build digital competence in research data management.
Participant Recruitment	- Recruited through university channels, summer/winter schools, SSH community newsletters, and DSIG Slack channel. - Focus on Early-Career Researchers (ECRs), data stewards, and PIs. - No prerequisite technical expertise required; focus on those working with or planning to work with audiovisual data.
Cost for Participants	- General-level course for PIs and support staff: ~€0-. - Extensive user course for ECRs and data stewards: ~€0. - Post-Grant Summer/Winter Schools: ~€400-€600 (includes meals and materials). Scholarships or fee waivers would be available for underfunded institutions.

Training Track:

The **Masking School & Website**, led by **Dr. James Trujillo**, **Babajide Owoyele**, and **Wim Pouw**, will provide targeted training programs and will prepare website materials:

- **General-level introduction course**: Two live/hybrid **1.5-hour sessions** aimed at PIs and support staff (**~20 participants per session**) at **VU University**.
- **Extensive user course**: Two **4-hour sessions** aimed at ECRs and data stewards (**~30 participants per session**) at **UvA** and **Rotterdam**.
- **Summer/Winter Schools**: Two 3-day events at **Radboud University** and **Donders Institute**, featuring Speed Talks from experts such as **Bogdana Huma** and **Inge Slouwerhof**.
- Recordings and training materials will be integrated into a **website for digital learning** for researchers who want to learn to apply masking.

Deliverables:

- Website offering training materials, FAQs, and recorded video tutorials.
- Real-time technical support platform.
- Certification for completing training modules.
- Pilot testing with Radboud Teacher Academy and technical report.

*Work Package 2: Platform Expansion Survey, Networking and Co-Creation, (Months 7-20)***WP 2 Overview:**

Track	Component	Target Audience	Who	What	Deliverables
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Training Track	Masking Lab	Extensive users, data stewards, ethics officers, infrastructure partners	Huma, Slouwerhof, Owoyele	One full-day workshop at VU, speed dating breakout sessions	FAQs, templates, and a policy research paper on challenges, bottlenecks, and opportunities.
Platform Track	SYNOPSIS Survey	Data stewards, researchers	RA(Technical), Owoyele	Survey on opportunities for upgrading the platform with gesture classification, prosody analysis, cloud and local use	Design requirements catalog for Improved platform, workflow guides, and policy paper

Objective: Survey enhanced platform functionality and co-creation forum for audio-visual masking with support from a broader community.



Figure 3 An Illustration of Masking (De-Identification and Facial plus Upper-Body Analytics), input frame (left), and masked frame (right) (video link: <https://osf.io/4ki6a/>).

Platform Track:

Babajide Owoyele and RA(Technical) will deploy a survey to explore opportunities to upgrade to a **cloud-based, FAIR-compliant system**. The survey will also explore how to support new enrichment features for audiovisual data, including text transcription, gesture classification, and prosody analysis. It will target both ECRs and support staff. We anticipate that TDCC SSH and DANS will help disseminate the survey. When a video is masked, as in Figure 3 above, analytics can be used to maximize the value of the masked data. We want to implement a Google survey to explore opportunities and challenges for integrating such analytics in future versions of the archiving platform; our survey targets ECRs and support staff and will be disseminated via newsletters from partner organizations and the DSIG Slack Channel & TDCC SSH Network via their website and newsletter (3000 plus individuals).

Component	Details
Target Audience	The Masking Lab is aimed at a diverse group of extensive users, data stewards, ethics officers, infrastructure partners, and researchers who deal with audiovisual data archiving, privacy issues, and FAIR workflows.

Audience Size and Composition	- Expected size: 30-40 participants. - Composition: • 10-15 data stewards from institutions involved in data archiving and management. • 5-8 ethics officers from universities or research institutions responsible for GDPR compliance and data privacy. • 5-8 infrastructure partners from organizations like SURF, DANS, and CLARIAH. • 8-10 early-career and senior researchers with experience or interest in working with audiovisual data.
Disciplinary Background of Participants	- Participants will come from diverse fields such as: • Digital Humanities, Linguistics, Social Sciences, Behavioral Sciences, and Cognitive Science. • Technical fields, including Computer Science, Data Science, and AI, especially for those involved in data archiving and tool development.
Facilitators and Instructors	Experts will lead the Masking Lab from both academic research and technical infrastructure, including: • Babajide Owoyele (DRIFT, EUR) – expertise in design science and responsible multimodal analytics. • Bogdana Huma (VU) – focus on communication linguistics, open science data, and ethics in data management. • Inge Slouwerhof (RU) – library sciences and data stewardship. • Infrastructure specialists from SURF and DANS for technical guidance on data storage, management, and GDPR compliance.
Course Objectives and Alignment	- Equip participants with practical knowledge of masking strategies and implementing these within institutional and ethical frameworks. - Co-create solutions for advanced masking challenges by leveraging collective expertise from ethics, data stewardship, and infrastructure. - Provide participants with tools and templates for embedding privacy-compliant workflows into their research projects and institutional protocols.
Recruitment of Participants	- Recruitment will be done via university networks (newsletters, department announcements), SSH community groups (DSIG Slack channel), and direct invitations to participants from partner institutions. • Ethics officers, data stewards, and infrastructure partners from collaborating institutions like SURF, DANS, and ODISSEI have been invited.

Training Track:

Bogdana Huma, Inge Slouwerhof, RA(Training), and Babajide Owoyele will organize the Masking Lab, which will serve as a co-creation forum for extensive users, data stewards, ethics officers, and infrastructure partners (**SURF, DANS, CLARIAH**). The forum will address ethical concerns and ensure FAIR-compliant workflows. It will also lead to the ideation of more advanced training modules to support audio-visual data archiving broadly.



Figure 4 Another Illustration of Masking (De-Identification and Facial plus Body Analytics), input frame (left), and masked frame (right) for a non-adult speaker.

Deliverables:

- A policy research paper on masking ethics and FAIR data management.
- Improved platform and workflow guides for researchers.
- Templates and co-created metadata standards for data stewards.

Work Package 3: Platform Rollout and Roadmap (Months 18-24)

WP 3 Overview:

Track	Component	Target Audience	Who	What	Deliverables
Training Track	Maskathon	Developers, data stewards	Pouw, Owoyele, RA Training	Collaborative event to explore opportunities to enhance de-identification benchmarks, platform features and ideate on improved research software support tailored to audio-visual data masking practices.	Certified training modules, improved masking algorithms
Platform Track	Platform Rollout	Dutch research institutions via SURF	van de Heuvel, Owoyele, RA(Technical)	Full-scale platform deployment, UI, and back-end optimizations	Final technical roadmap benchmarks for masking tools

Objective: Complete development and ensure sustainable support for SYNOPSIS while creating a roadmap for future growth.

Platform Track:

Led by **Babajide Owoyele** and **Research Assistant (Technical)**, the platform will scale from the pilot stage to full deployment across several Dutch research institutions via **SURF**. This phase will focus on optimizing the user interface and back-end performance.

Training Track:

RA (Training), **Wim Pouw**, and **Babajide Owoyele** will organize a **Maskathon**. The event will foster collaboration

between developers, data stewards, and computer scientists to ideate benchmarks for masking and introduce new algorithms to support continuously improved masking, such as in computer science/machine learning communities. Additionally, **data stewards at partner universities** will ideate on a train-the-trainer workshop in **Netherlands eScience Center**.

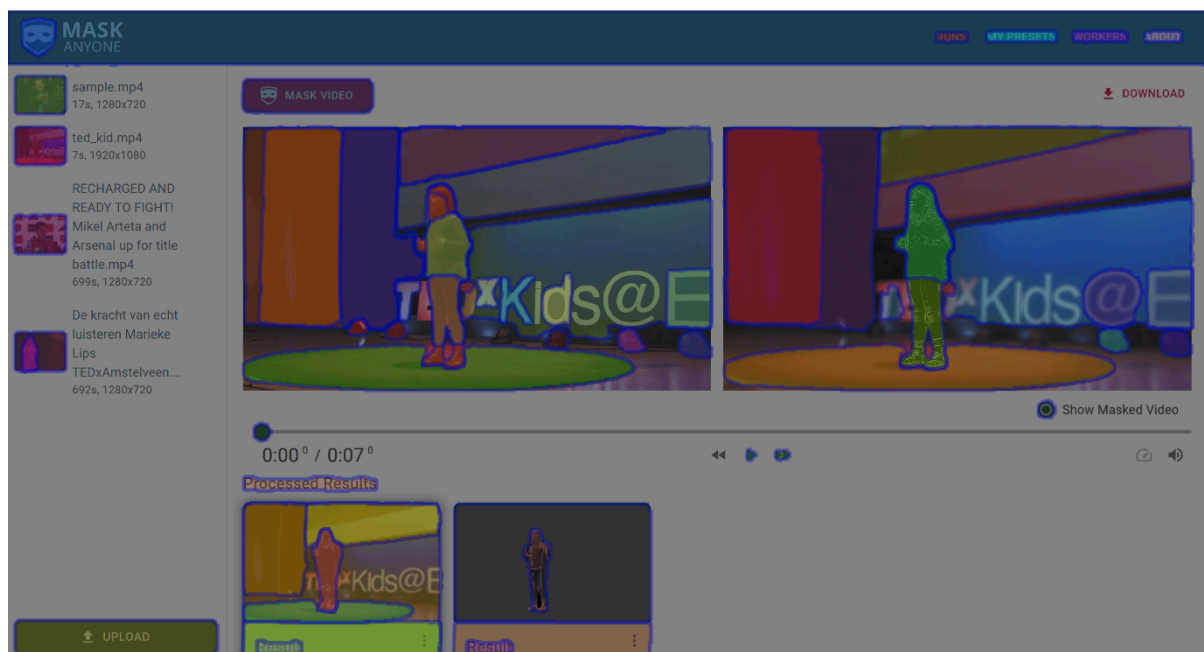


Figure 5 Maskathon - A Hackathon-style event to ideate new benchmarks and trainings for MaskAnyone (User Interaction and The CodeBase)

Roadmap:

A comprehensive roadmap detailing benchmarks, future developments, and sustainability strategies will be published, including potential funding from **NLeSC** and **Open Science NL Infrastructure**.

Deliverables:

- A final technical roadmap paper.
- Benchmarks for evaluating masking tools.
- Prototype of a Certified training module for developers and data stewards.

3. Declarations and signature

Funding elsewhere

Have you requested funding for this project elsewhere?

- ☒ No
☐ Yes

Please include details of any additional grants you have requested for this project.

By submitting this form, I declare that:

- ☒ I have completed this form truthfully;
☒ I meet the nationally and internationally accepted standards for scientific conduct as stated in the Netherlands Code of Conduct for Research Integrity 2018;